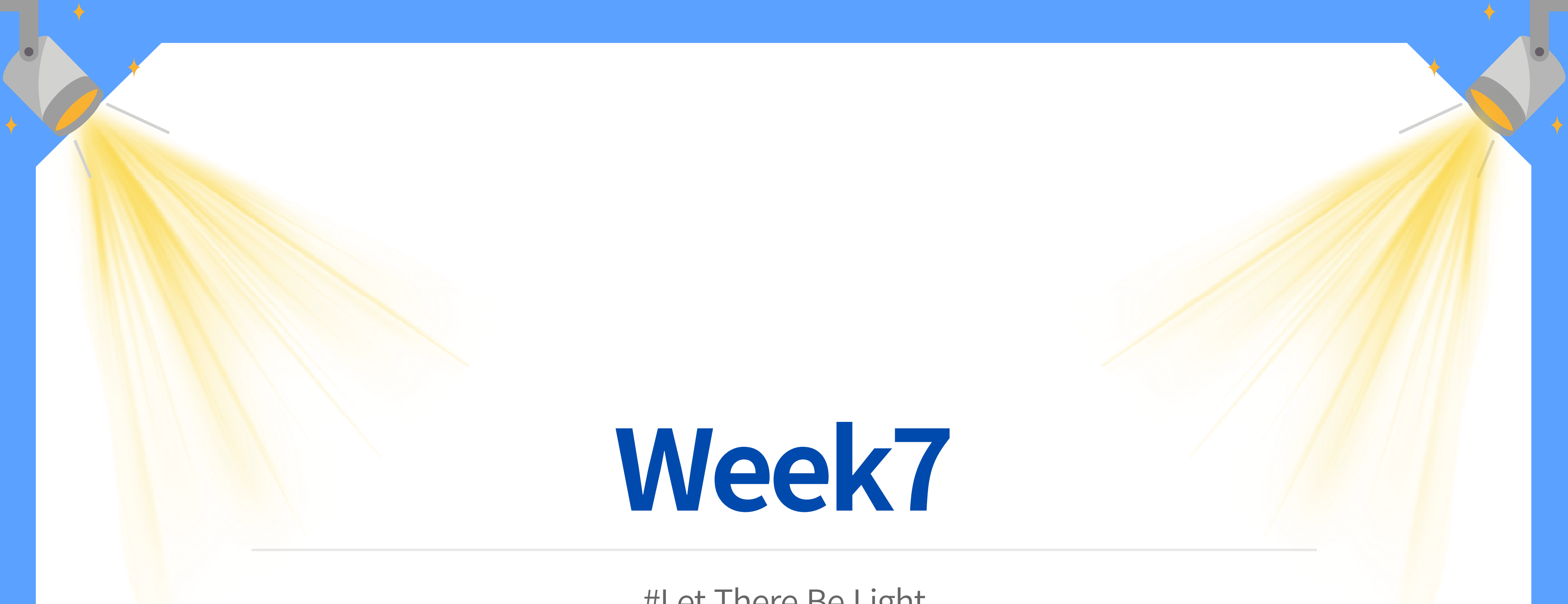




Week7

#Let There Be Light

Shading / Lighting Model

Per-Vertex

Gouraud



Per-Pixel

Lambert



$N \cdot L$ 로 **Diffuse** 계산

Blinn-phong



Diffuse + Specular
H를 이용해 Specular 계산

N: 표면의 법선 벡터 **L**: 광원 방향 벡터 **H**: $L + \text{ViewDirection}$

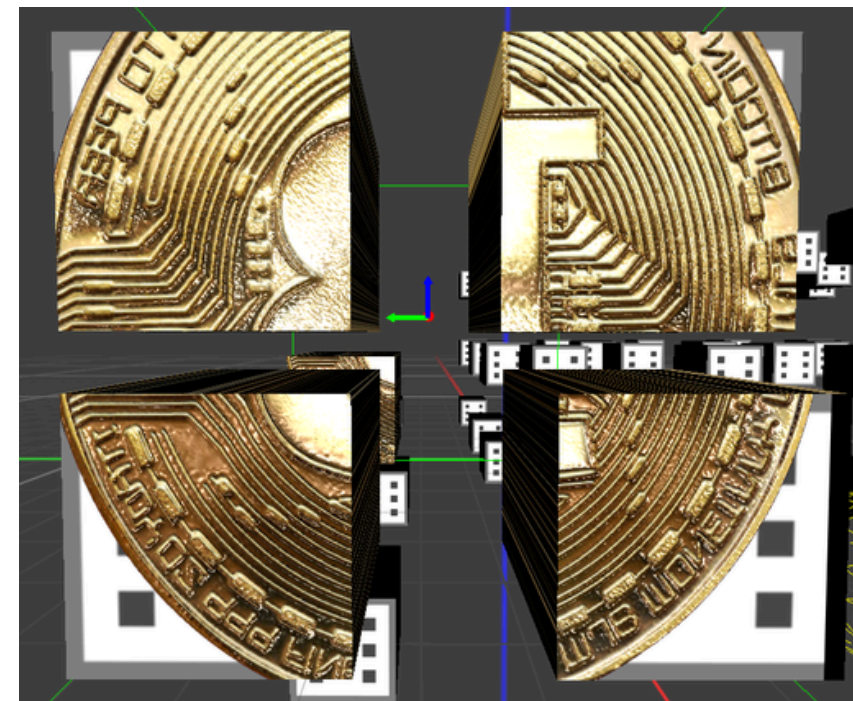
Light to Decal

WorldNormal(투영된 Mesh Normal) 기반



Decal Material의 고유 입체감 표현 불가능

Decal NormalMap 기반

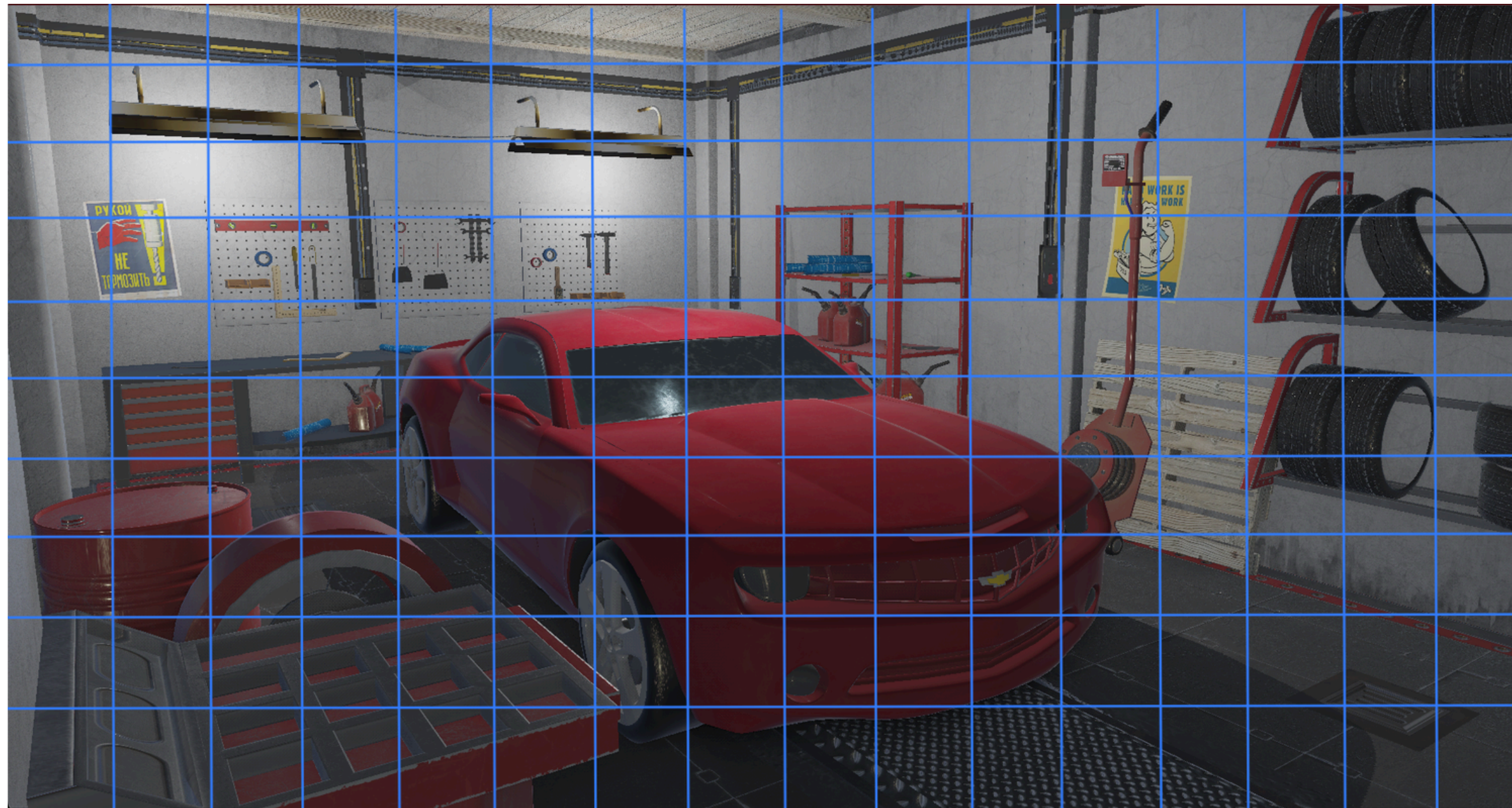


Mesh의 굴곡과 Decal의 입체감 표현 가능

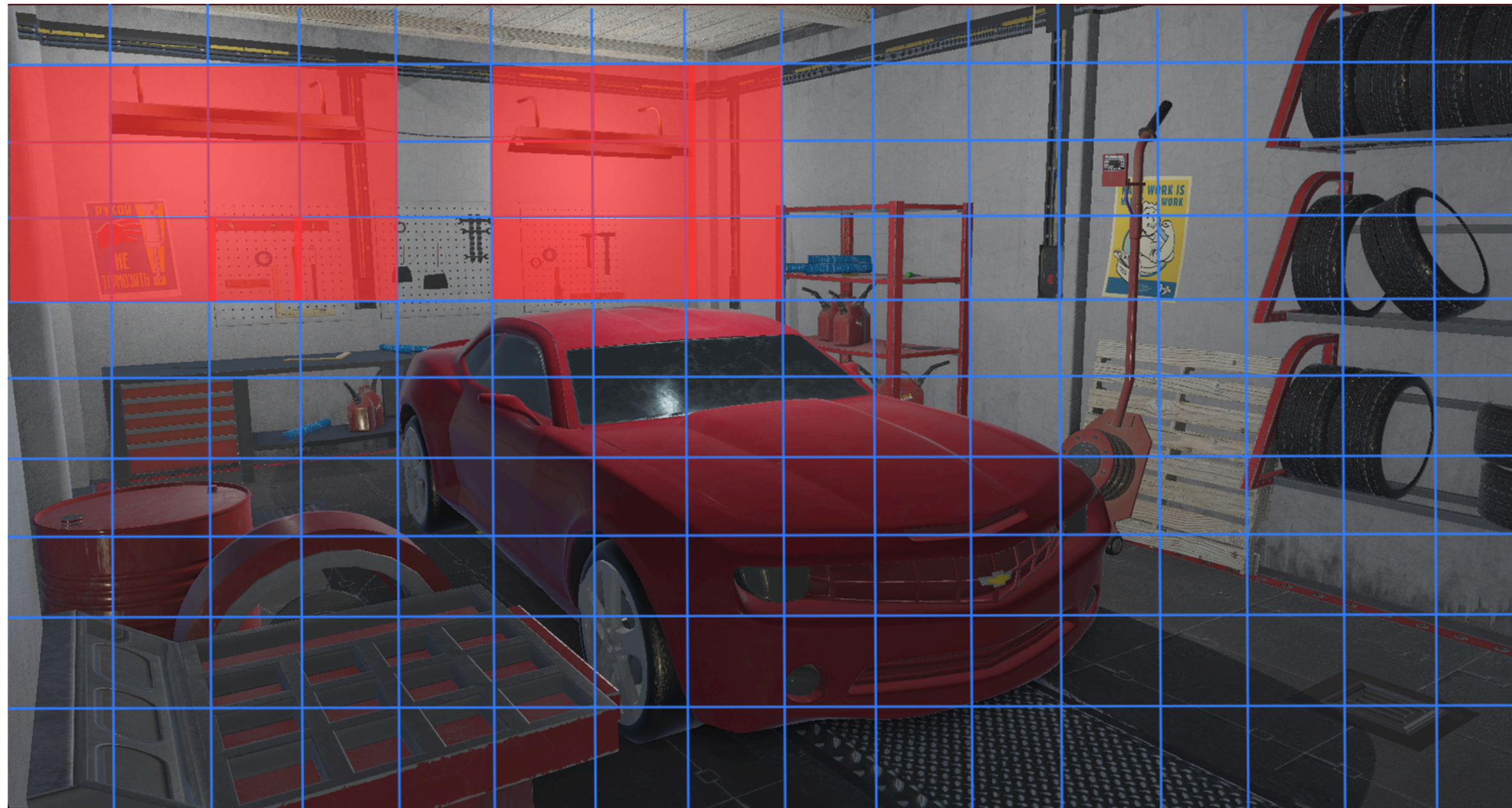
2.5D Tile Based Light Culling

“타일별 깊이값을 활용해, 실제로 영향 있는 광원만 남긴다!”

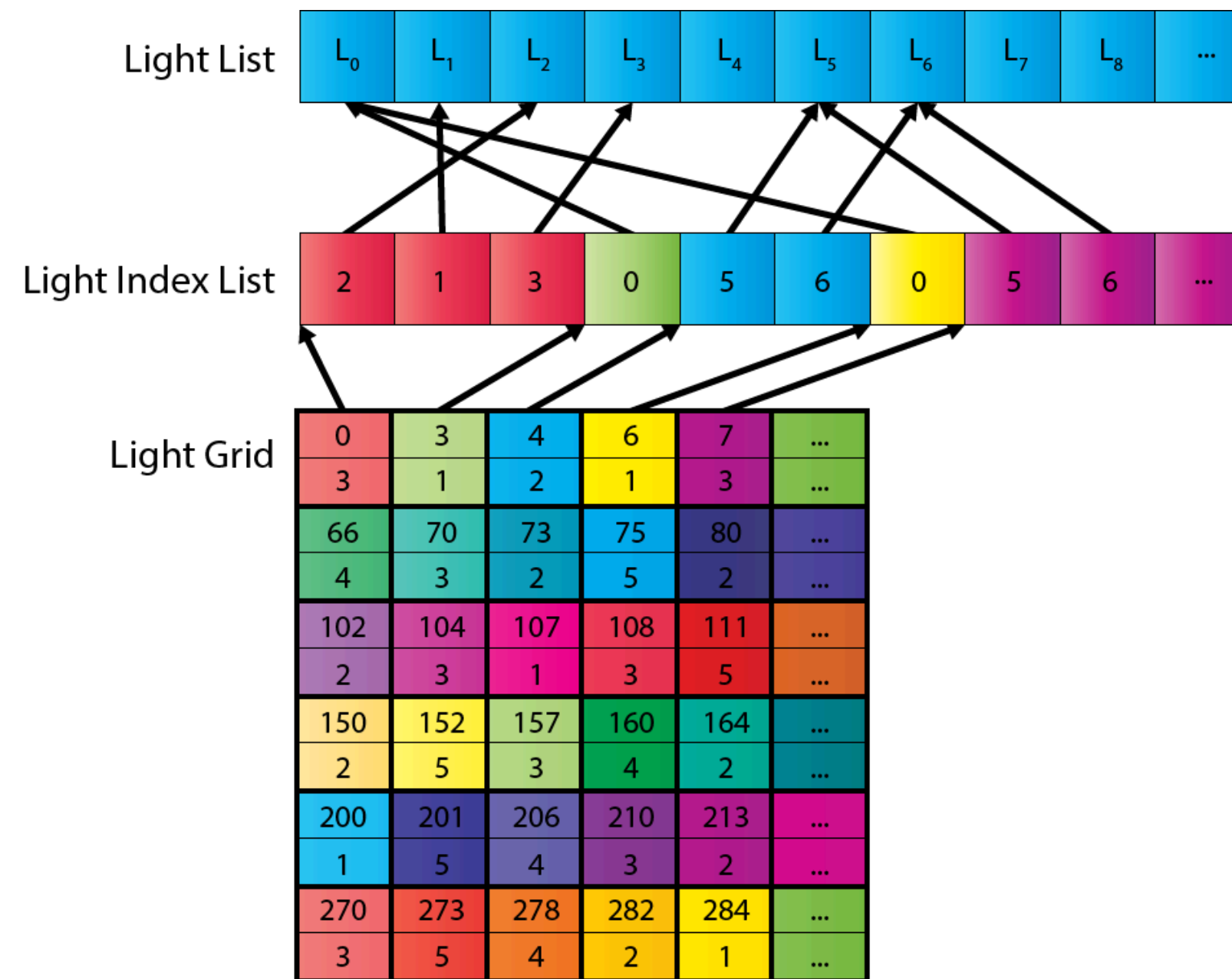
Light Culling



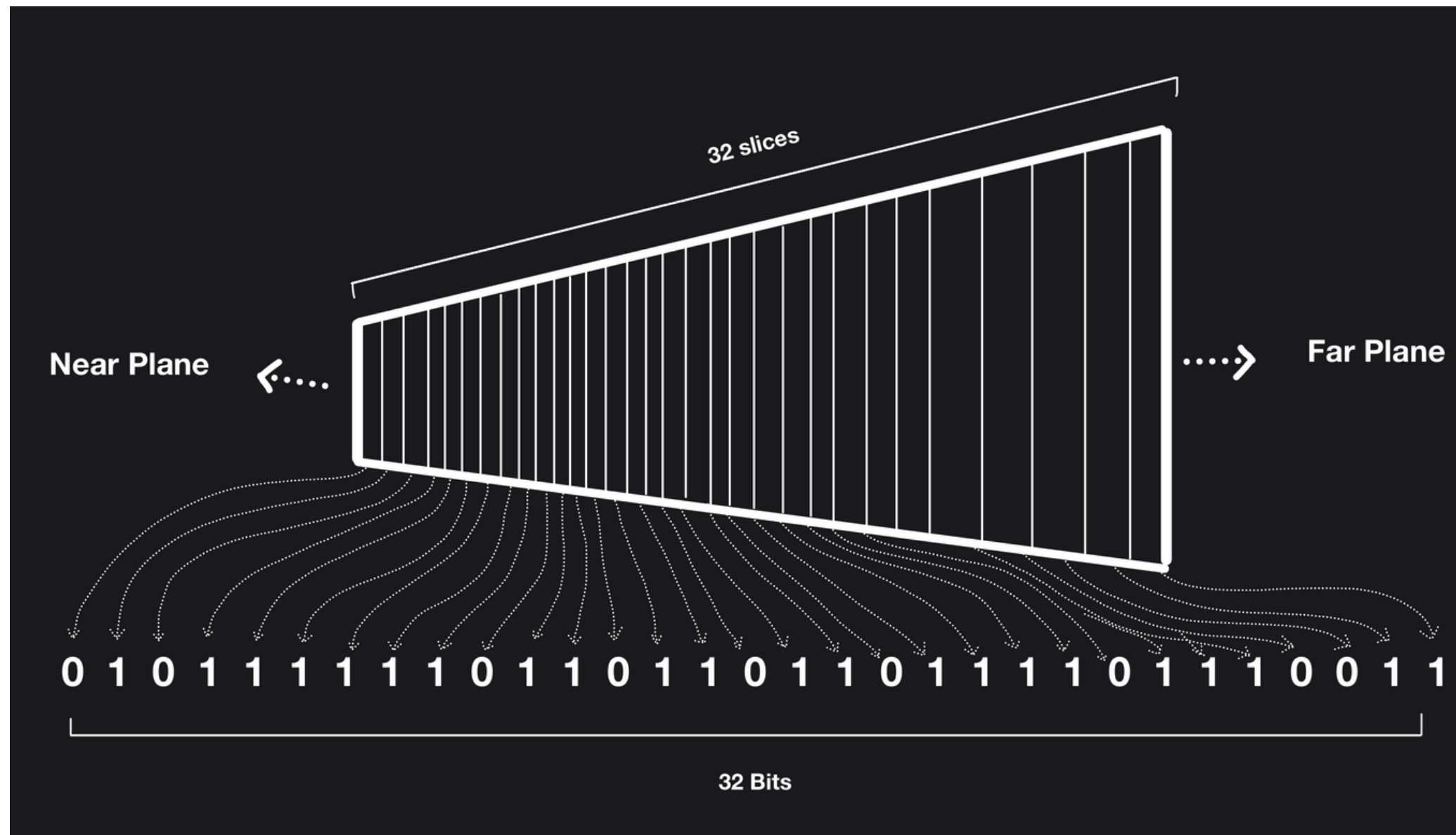
Light Culling



Light Culling

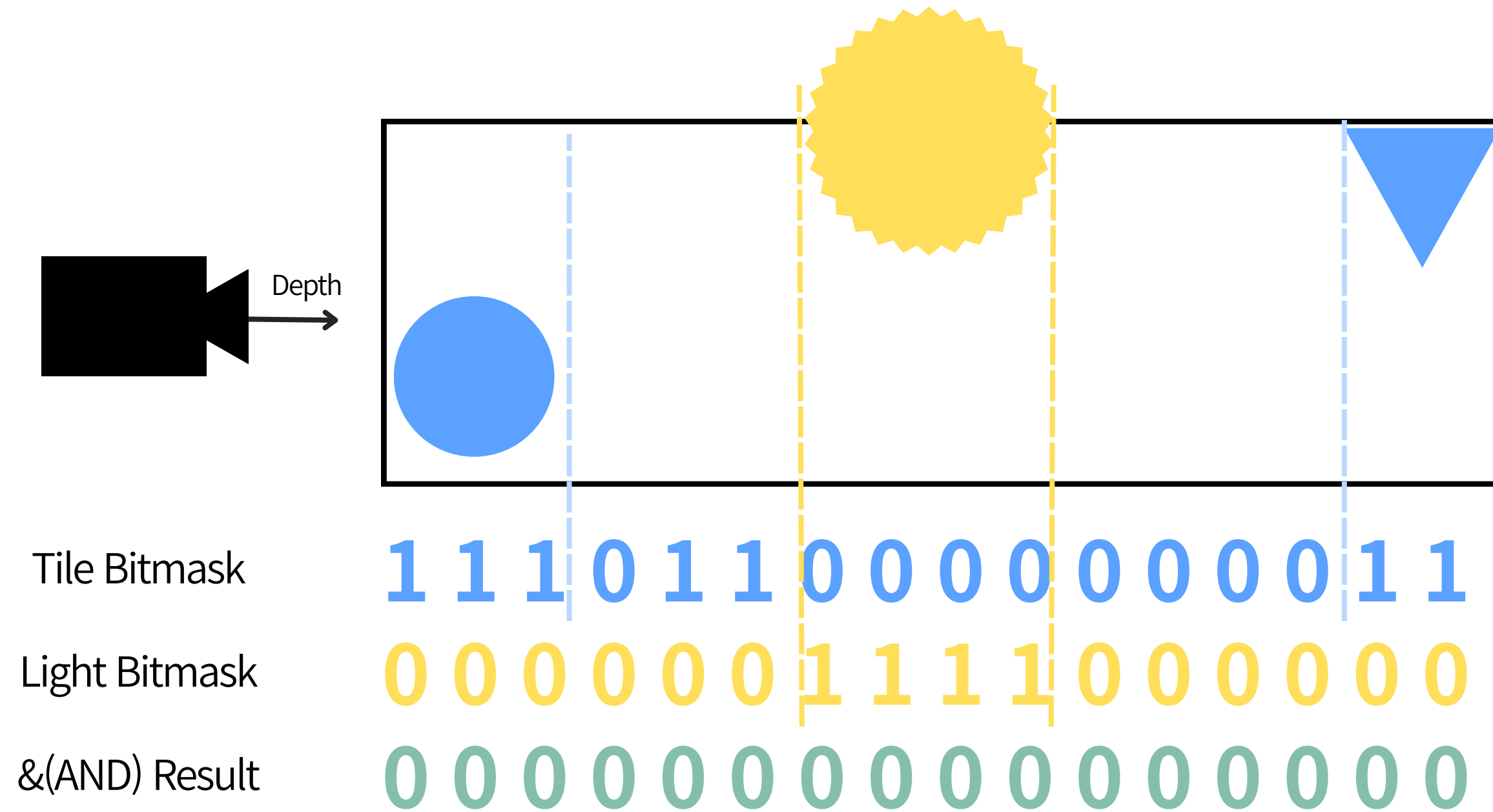


Light Culling



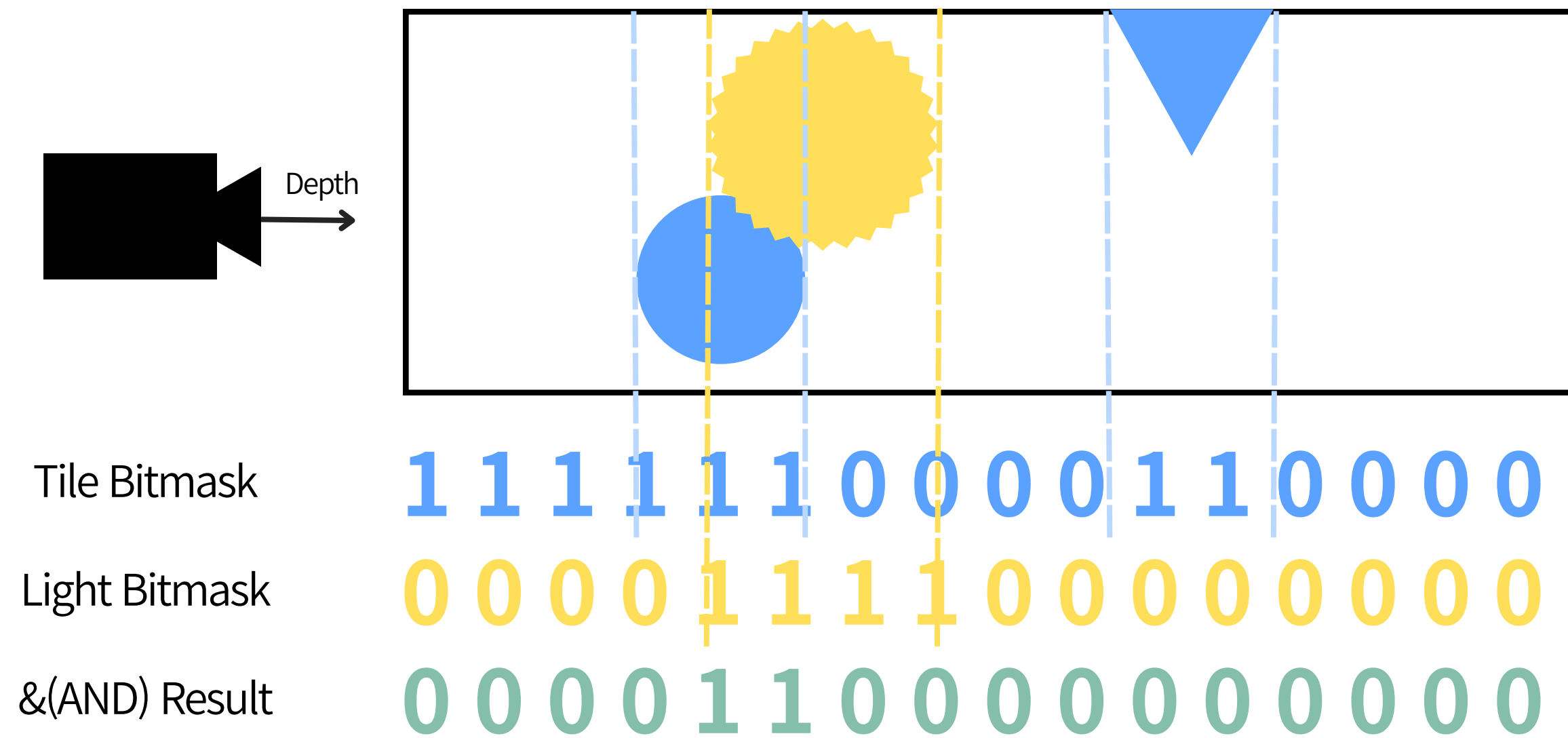
Light Culling

case : bitmask non-overlap

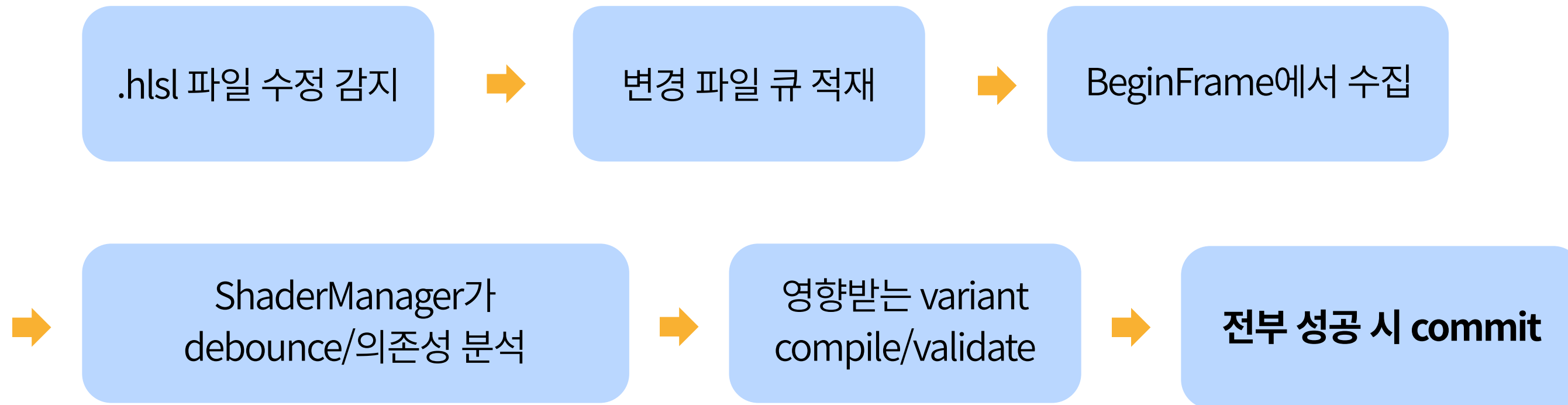


Light Culling

case : bitmask overlap



Shader Hot Reload



- ReadDirectoryChangesW 활용

엔진 시연



발표를 들어주셔서
감사합니다

THANK YOU

Team 7

김형도 | 박세영 | 조현석